A Survey of LLM×DATA

2.3 Data Processing for LLM

For CENG544

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Introduction to 2.3 Data Processing for LLM

Large Language Models (LLMs) learn by reading enormous amounts of data (text, code, images, etc.).

However, raw data from the real world is messy: it is duplicated, noisy, biased, incomplete, or unsafe.

Data processing for LLMs is the systematic pipeline that:

- collects data,
- cleans and organizes it,
- improves its quality,
- and prepares it

so that an LLM can learn effectively and safely.

2.3.1 Data Acquisition

What it is: Process of collecting raw data that an LLM will learn from

Key Principle: Unlike traditional machine learning (which uses small, labeled datasets), LLMs need massive amounts of diverse data from across the internet

Data Sources:

1. Public Data: Freely available information
2. Private Data: Confidential information requiring special handling

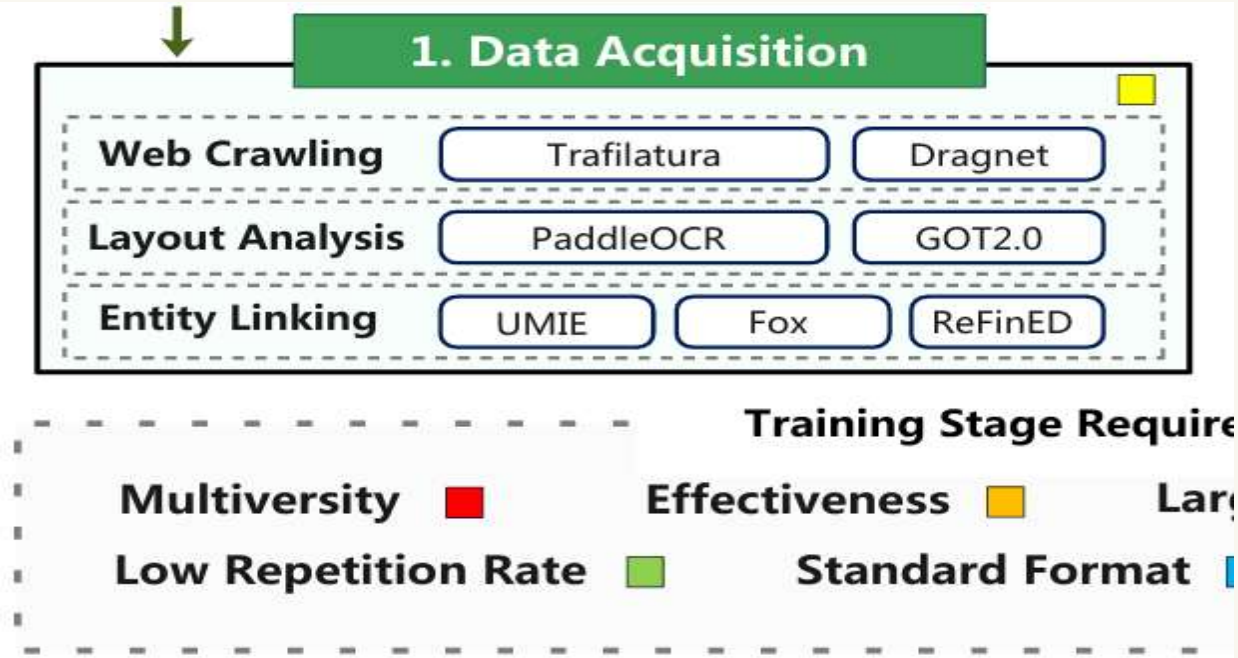
2.3.1 Data Acquisition

Data Acquisition Methods:

A. Website Crawling
- Rule-based Crawling
- ML-based Crawling
- Auxiliary Tools

B. Layout Analysis
- Pipeline-based Methods
- End-to-End Models

C. Entity Recognition & Linking



2.3.2 Data Deduplication

<u>What it is:</u> Removing repeated or near-identical data samples

<u>Why it matters:</u> Duplicates can make models overfit and waste training resources

<u>Four Main Categories:</u>

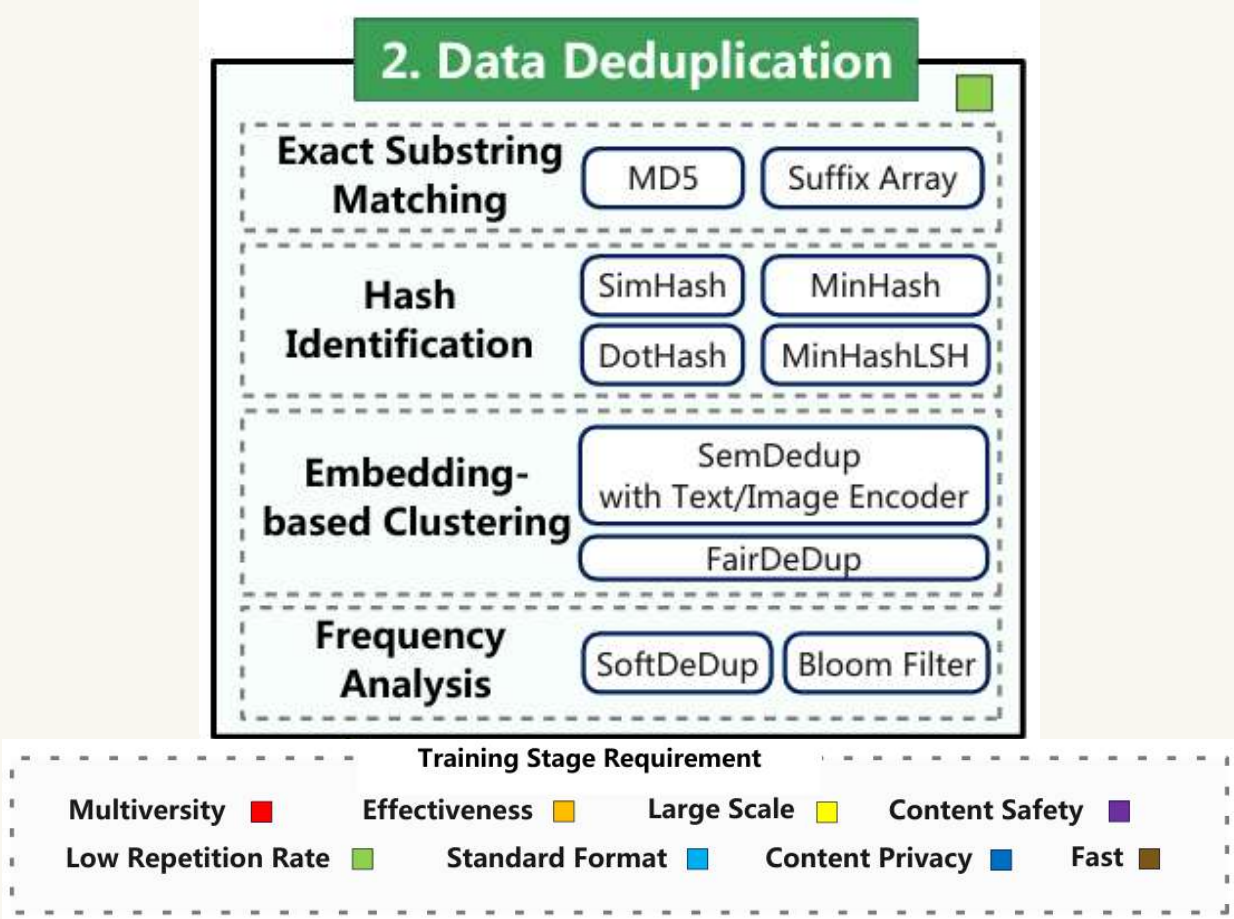
A. Exact Substring
- MD5 Hashing
- Suffix Array

B. Approximate Hashing-based Deduplication
- MinHash
- SimHash
- DotHash

C. Frequency-based Down-Weighting
- SoftDeDup

D. Embedding-Based Clustering
- SemDeDup
- FairDeDup

2.3.2 Data Deduplication



2.3.3 Data Filtering

<u>What it is:</u> Removing low-quality or harmful samples

<u>Goal:</u> Keep only high-quality, safe samples that cover various topics

<u>Two Main Levels:</u>

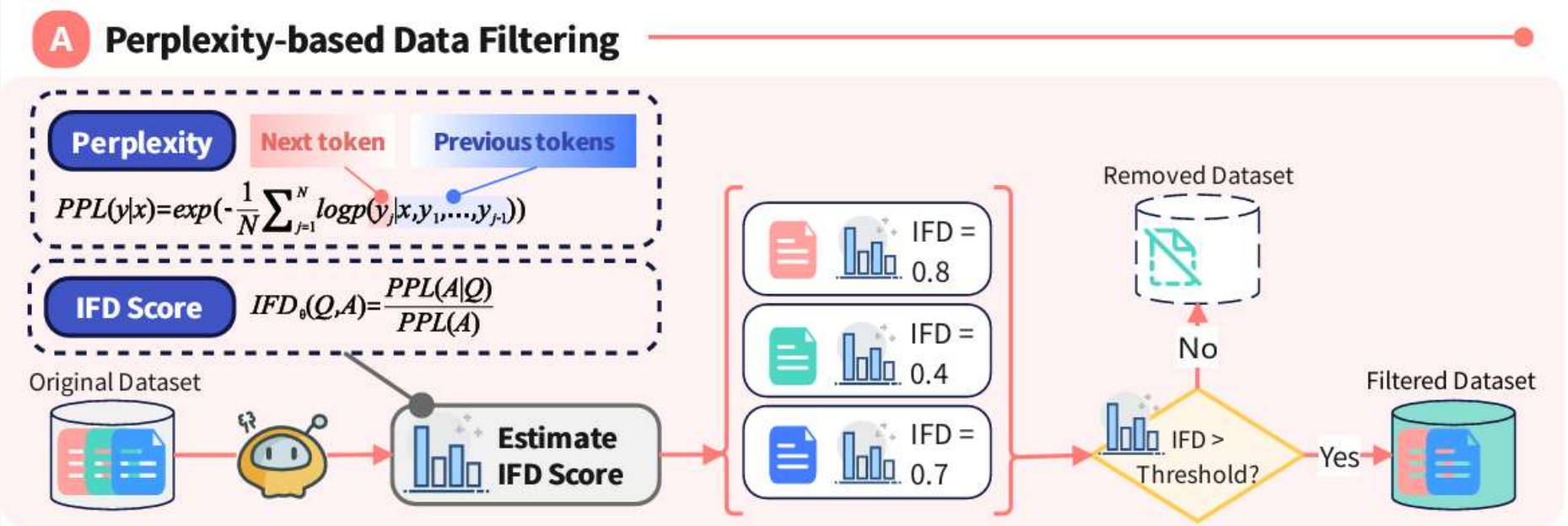
A. Sample-Level Filtering

- Statistical Evaluation
  - Perplexity Measuring
  - Influence Assessment
  - Clustering
- Model Scoring
- Hybrid Methods

B. Content-Level Filtering

- Privacy Anonymization
- Image & Video Filtering

2.3.3 Data Filtering



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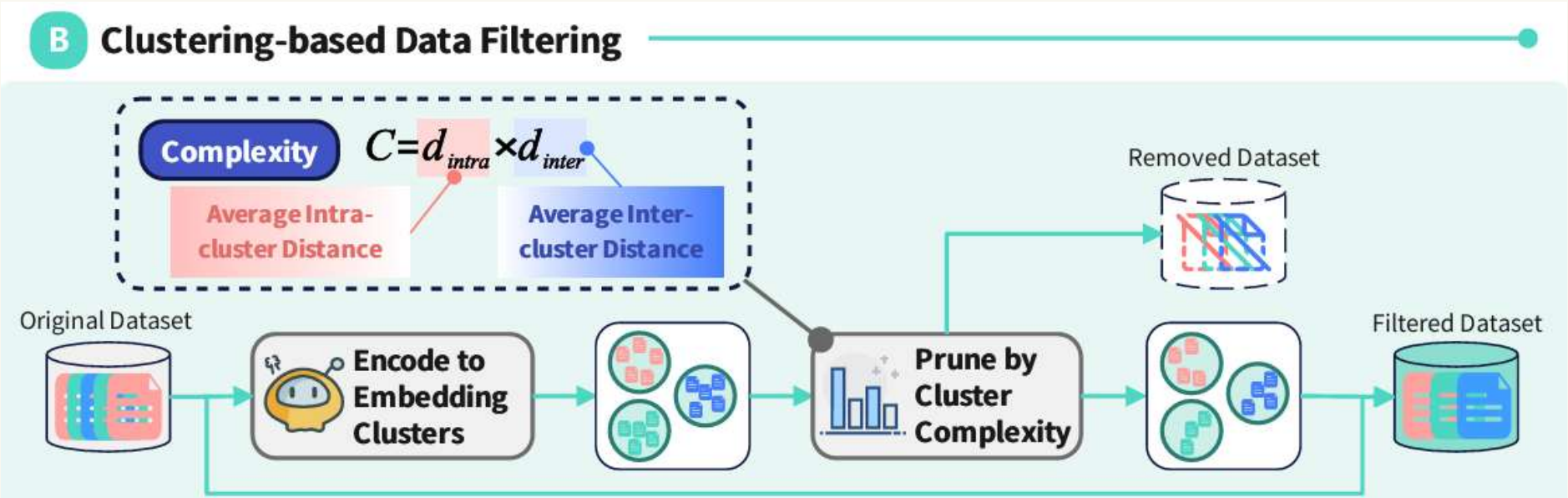

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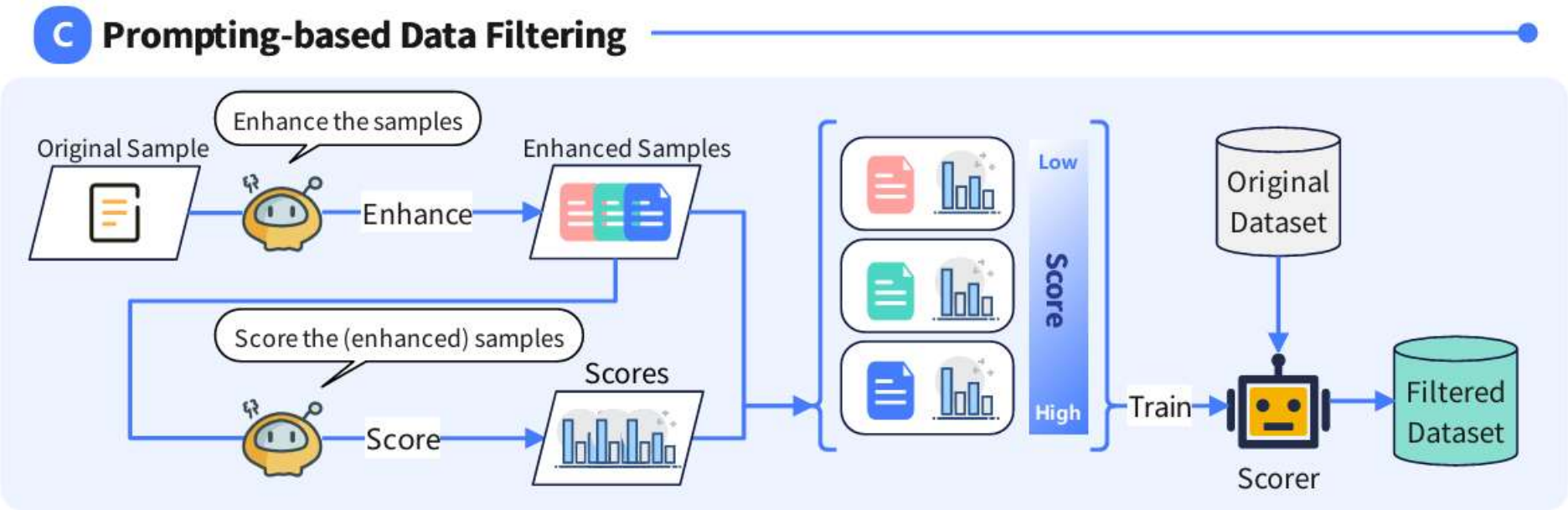

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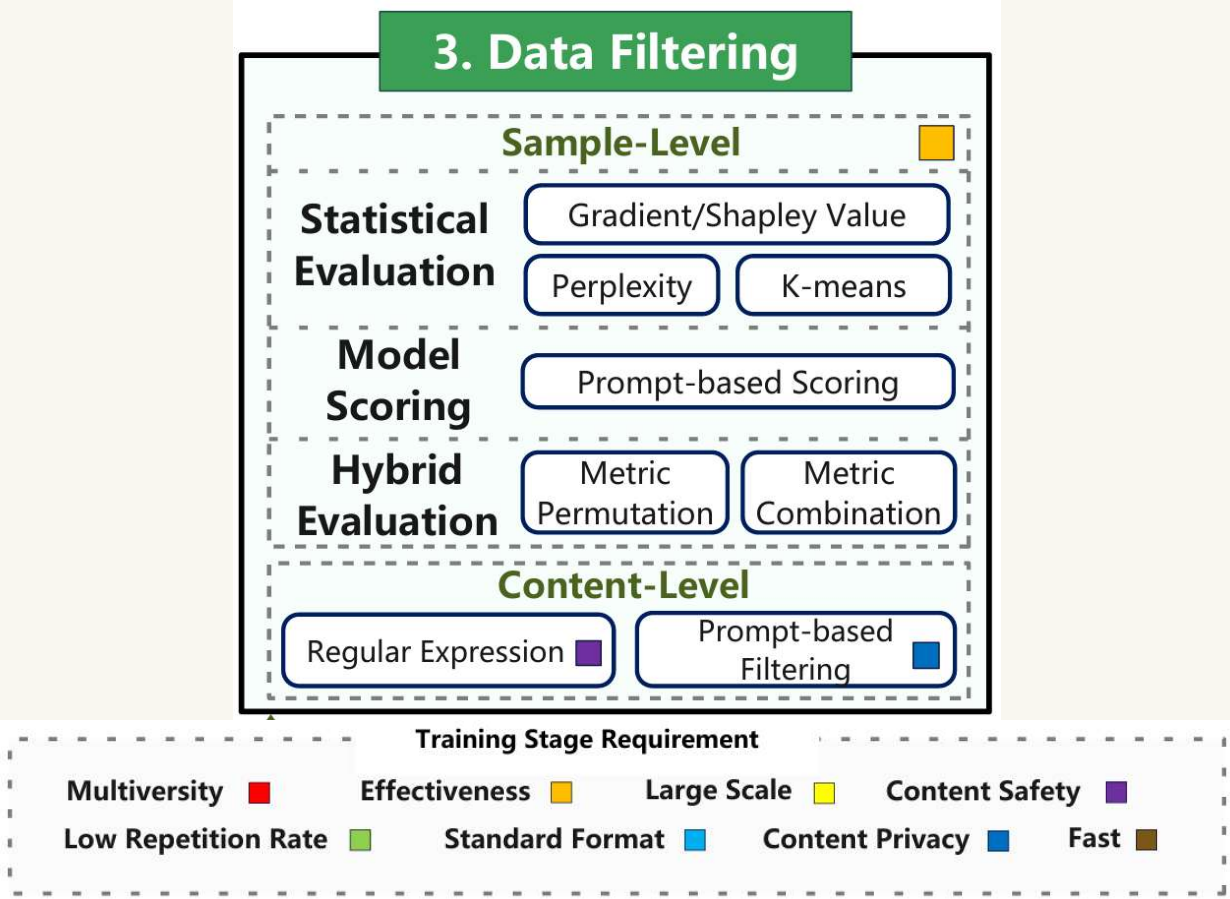

2.3.4 Data Selection

<u>What it is:</u> Choosing subsets of data that best represent the target distribution

<u>Goal:</u> Select the most relevant data for specific tasks or domains

<u>Three Approaches:</u>

A. Similarity-Based Selection

- Cosine Similarity
- Bag-of-Words Similarity
- Lexicon Set Overlap
- Bayes-based (CoLoR-filter)

B. Optimization-Based Selection

- Linear Search (DsDm)
- Gradient-based Search
- Kernel Density Regularization

C. Model-Based Selection

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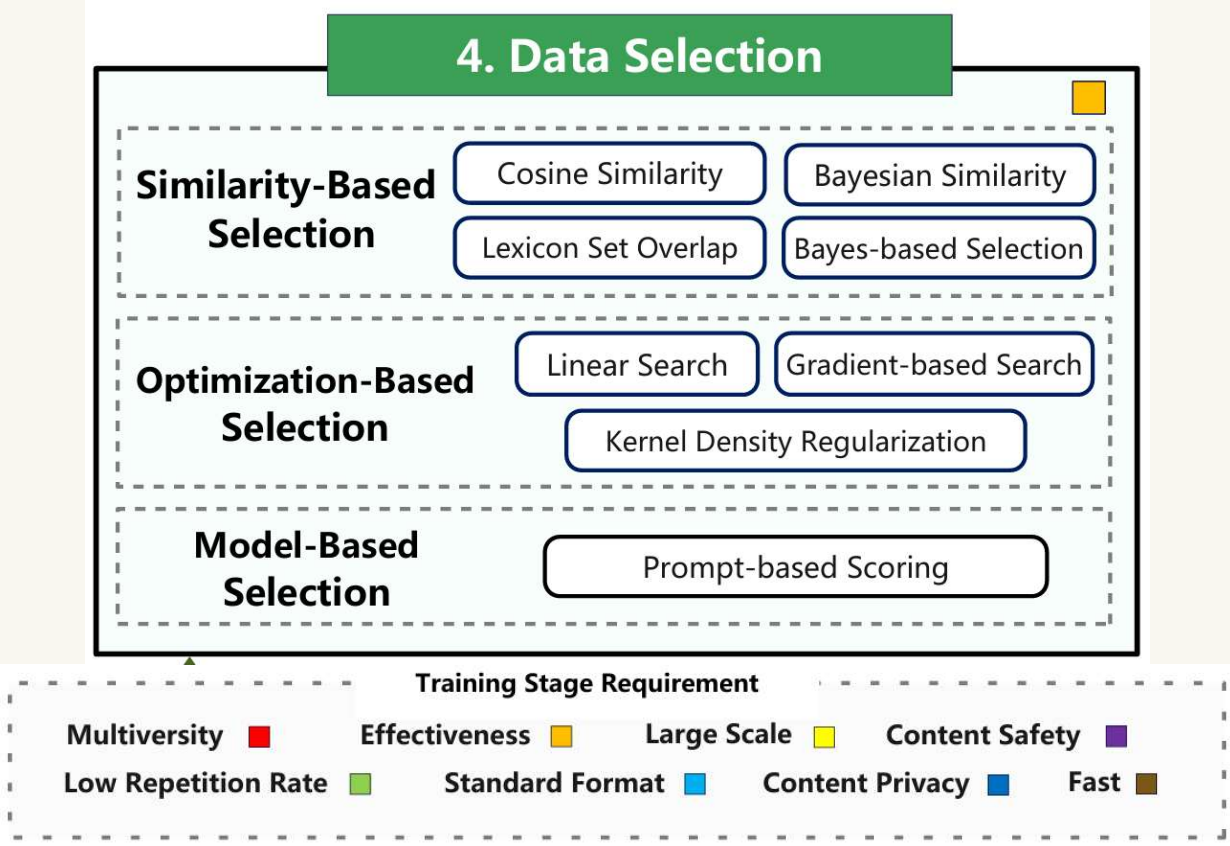

2.3.5 Data Mixing

<u>What it is:</u> Combining datasets from different domains in specific ratios.

<u>Goal:</u> Preventing overfitting and ensuring broad capabilities

<u>Two Main Levels:</u>

A. Before-Training

- Human Experience
- Model-Based Optimization

B. During-Training (Dynamic)

- Bilevel Optimization
- Distributionally Robust Optimization
  - Group DRO
  - Task-level DRO

2.3.5 Data Mixing: Before-Training

Taxonomy	Stage	Methods	Traits
Before Training (Human Experience)	Pre-training	Multi-Source Data Adjusting [139], [347]	Intuitive and easy to implement, suitable for rapid experimentation.
		Entropy-Based Mixing [152]	Low computation cost with quality quantification by entropy.
Before Training (Model-Based Optimization)	Pre-training	Linear Regression Model [263]	Only 10% of DoReMi's [420] computational resources are required. Simultaneously train hundreds of small models to accelerate optimization.
	Pre-training	Bivariate Data Mixing Law [152]	Avoid iterative training of proxy models (low computational costs). Show relation between loss and training steps.
	Continual Pre-training	Chinchilla Scaling Law [323]	Support knowledge transferring to new domains (↓ over 95% training costs).
	Pre-training	Exponential Functions [439]	Support datasets without explicit domain division.
	Continual Pre-training	Power-law Function [160]	Compared to single-objective optimization like [323] [160] ensures that domain performance improvement does not compromise general capabilities.
	Pre-training	Classification Model [251]	Reverse engineering for finding the suitable data recipe of LLMs.
During Training (Bilevel Optimization)	Pre-training	Calculate domain contribution by gradient inner products [135]	Requires a proxy model, performances well in OOD datasets.
	Fine-tuning	Dynamically adjust weights by gradient alignment values [302]	Multiple applications like multilingual training, instruction following, large-scale data reweighting
During Training (Distributionally Robust Optimization)	Pre-training	Group DRO [420]	For pre-training, smooth adjusting to prevent abrupt weight changes
	Fine-tuning	Task-level DRO [278]	For fine tuning, quick response to task difficulty changes

2.3.6 Data Distillation and Synthesis

- We are running out of high-quality human data.
- Use LLMs (Teachers) to generate training data for other LLMs (Students).
  - Data Scarcity
  - Privacy Concerns
  - Data Diversity
  - Reasoning Data
  - Human Alignment

2.3.6 Knowledge Distillation

- Transferring capabilities from Strong Models (e.g., GPT-4) to Weak Models.
- Task-Specific Prompting
  - Chain-of-Density
  - Force teacher LLM for more detailed summarizations of the text
- Program-Aided Distillation
  - Using a program – i.e python interpreter – to verify teacher output before training student LLM.

2.3.6 Pre-training Augmentation

- Injecting diversity into the massive base corpus.
- Rephrasing (WRAP)
  - rephrase the same text in different styles to keep facts while reducing overfitting.
- Domain Synthesis
  - JiuZhang 3.0 (Math)
  - Case2Code (Unit Tests for Code)

2.3.6 Reasoning & Verification

- Reasoning: The model is encouraged to generate step-by-step logic instead of only final answers.
- Verification: Generated answers are checked using rules, external tools, or reference answers.
- Robustness: Outputs are tested with slightly changed or harder inputs to see if reasoning stays consistent.
- Limitations: Reasoning can still look correct while being wrong; verification reduces but doesn't eliminate this.

2.3.7 End-to-End Data Processing Pipelines

- Unified workflow: raw data → training data
- Designed as a scalable system
- Supports TB-scale processing
- Enables reproducibility

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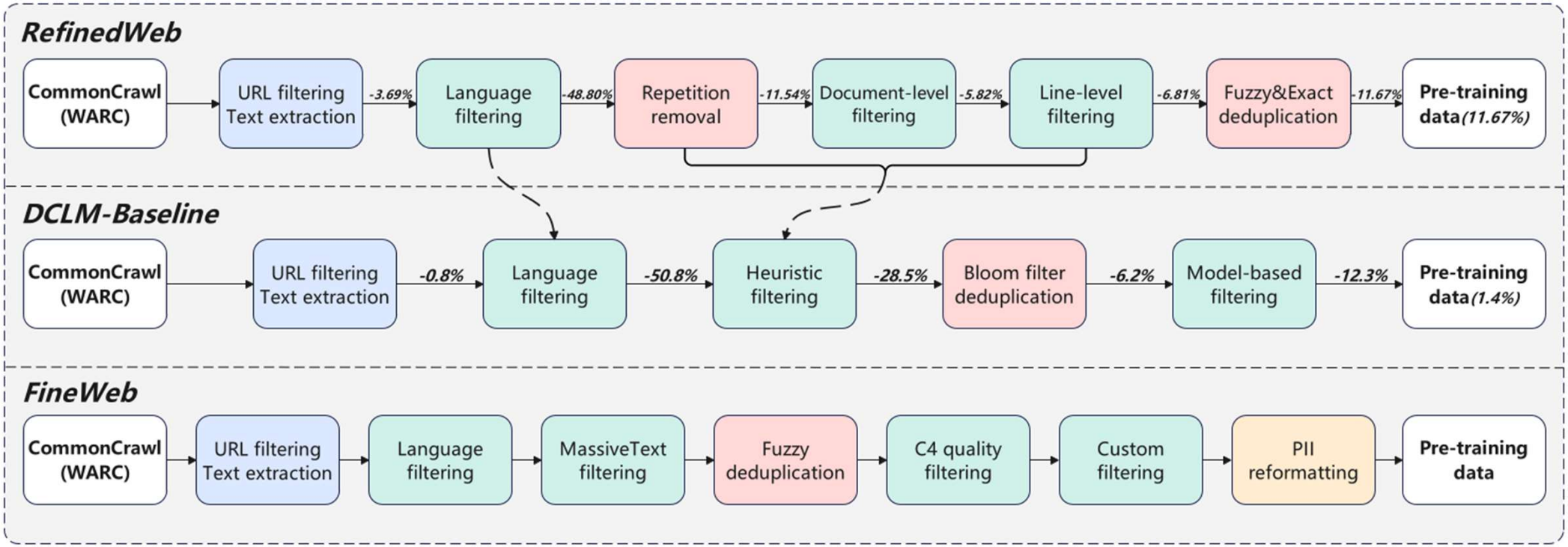

Fig. 6: Typical data processing pipelines for LLMs.

Questions?